

Srishti Lamba

M.Sc. Data Science

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Education

Dhirubhai Ambani University (DAU)

M.Sc. Data Science | CGPA: 8.6 / 10

Jul 2025 – Present

Gandhinagar, Gujarat

- Coursework: Machine Learning, NLP, Statistical Modelling, Big Data Analytics, Causal Inference, Interactive Data Visualization

Maharaja Sayajirao University of Baroda

B.Sc. Statistics (Honours)

Jun 2021 – May 2024

Vadodara, Gujarat

- Coursework: Probability Theory, Regression Analysis, Time Series Analysis, Design of Experiments, R Programming

Projects

RupeeRisk | India Macro & Geopolitical Forex Intelligence Platform | GitHub · Streamlit

Feb 2026 – May 2026

- Engineered 15 geopolitical tension indicators across 4 transmission channels (Oil Supply, Direct FX, Risk-Off, Trade Route) using STL decomposition, ADF unit-root diagnostics, and 1-week exogenous lags to eliminate look-ahead bias; verified stationarity of all series (USD/INR ADF = -7.64 , $p < 0.0001$).
- Built a 9-model rolling backtest pipeline (ARIMA, ARIMAX, Lasso, Gradient Boosting, Random Forest, 2 hybrids, Naïve Random Walk, SES) over a **200-week out-of-sample window**; best model (ARIMA+Lasso Hybrid) achieved **0.332% MAPE**, **59% Directional Accuracy**, Sharpe 1.33, and **+18.07% cumulative return**.
- Replaced seasonal SARIMA with ARIMA(1,1,0) after ACF/PACF audit confirmed no significant seasonality at lags 13, 26, or 52; benchmarked all models against naïve random walk using Theil's U.
- **Stack:** Python · yfinance · FRED API · statsmodels · pmdarima · scikit-learn · Streamlit · Plotly

IndiaShield | Strategic Defence Intelligence & Analytics Platform | GitHub · Streamlit · Tableau

May 2026 – Present

- Built a 6-module strategic intelligence dashboard consolidating 25 years of SIPRI military expenditure data, Union Budget allocations (FY16–FY25), SIPRI arms transfer TIV flows, and NSE defence equity prices across India, China, Pakistan, and the USA.
- Engineered the **SIDS (Strategic Import Dependency Score)** — a novel composite metric quantifying India's arms supply-chain vulnerability per supplier using import concentration, single-source risk, geopolitical stability, and domestic substitution capacity; classified Russia at SIDS = 72 (Critical) post-2022 sanctions.
- Quantified the **Modernisation Paradox**: CapEx share fell 38.3% (FY16) → 28.9% (FY25) despite $2.5\times$ nominal budget growth; China's defence spending grew $9.6\times$ vs India's $5.4\times$ (constant 2022 USD), widening the gap from \$17 Bn to \$232 Bn.
- Event study on 6 NSE defence equities across 9 geopolitical escalations (Uri, Balakot, Galwan): HAL and BEL generated positive cumulative alpha over a ± 30 trading-day window while Nifty 50 declined.
- **Stack:** Python · Streamlit · Plotly · Tableau Public · SIPRI MILEX/TIV · yfinance · pandas · NumPy

ShramikSaathi | Indian Worker Rights AI Copilot | GitHub | Team of 4

Jan 2026 – Apr 2026

- Co-built a 9-layer RAG + ReAct pipeline on LLaMA 3.1 8B covering PF/EPFO, payslip audit, labour rights, and income tax; stages: cross-domain router, slot extractor, sufficiency gate, FAISS retriever (311 docs), procedural eligibility reasoner, DPO-aligned generator.
- Contributed the novel **Condition Coverage Score (CCS)** — measures whether eligibility conditions are verified *before* generation; CCS improved from **0.000** → **0.714** via LoRA SFT + DPO alignment ($\beta = 0.05$); sufficiency gate accuracy: $0.500 \rightarrow 0.991$.
- DPO training: 373 preference pairs across 4 dimensions (grounding, verdict correctness, citation discipline, refusal/escalation); $\beta = 0.05$ achieved grounded_clean = 0.750 with zero citation fabrication on held-out set.
- **Stack:** LLaMA 3.1 8B · LoRA (PEFT) · DPO (TRL) · FAISS · sentence-transformers · Gradio · Lightning AI A100

LogiSense | Logistics Intelligence & Analytics Platform | GitHub · Live Demo | Team of 3

Aug 2025 – Nov 2025

- Designed a 7-table BCNF-normalised PostgreSQL schema for logistics operations; authored 20+ analytical SQL queries across 9 dashboard tabs covering branch failure analysis, IQR-based charge outlier detection, payment mismatch detection, and percentile-based customer segmentation (VIP / Premium / Regular / New / Inactive).
- Built a zero-backend HTML/JS analytics dashboard (Netlify-deployed) with demo mode and live SQL query viewer; adopted for pilot evaluation by 2 logistics companies.
- **Stack:** PostgreSQL · SQL · Python · pandas · HTML/CSS/JS · Netlify

A World in Numbers | *Global Inequality, Health & the Planet* | Team of 4

Feb 2026 – Apr 2026

- Led statistical analysis across 195 countries; built dual-axis Tableau visualisations, applied DAG causal inference, and identified Out-of-Pocket health expenditure as a confounder masking the tax-health relationship ($R^2 = 0.13$); proposed a policy framework for equitable health financing.
- **Stack:** Python · Tableau · Hypothesis Testing · DAG Causal Inference · WHO GHE-D · Global Carbon Project

Screen Time & Academic Performance | *Minor Research Project, DAU*

May 2026 – Present

- Conducting a 3-wave longitudinal survey ($n = 30\text{--}35$) at DAU; applying Bland-Altman analysis to quantify self-reported screen time bias and building a predictive model to quantify its effect on academic performance and mental wellbeing.
- **Stack:** Python · R · OLS Regression · Bland-Altman Analysis · DAG Causal Inference · Predictive Modelling

Technical Skills

Languages & Tools: Python, R, SQL, PostgreSQL, Excel, Git/GitHub, Jupyter Notebook, statsmodels, pmdarima, scikit-learn

BI & Visualization: Tableau, Power BI, Streamlit, Plotly, Matplotlib, Seaborn

ML & Analytics: Lasso, Gradient Boosting, Random Forest, ARIMA/ARIMAX, LoRA, DPO, FAISS, Feature Engineering, Customer Segmentation

Statistics: Hypothesis Testing, OLS/Multivariate Regression, Causal Inference (DAGs), STL Decomposition, ADF Testing, Bland-Altman Analysis, Time Series Analysis

Certifications

Google Business Intelligence Certificate | Google | 2026

Product Analytics Certification | Pendo & Mind the Product | 2026

Databricks Data Analyst Certification | Databricks | *In Progress*

Achievements & Volunteering

Defence Examinations: Cleared AFCAT & CDS Written Examinations and progressed to SSB selection for Indian Air Force Commission — demonstrating discipline, leadership under pressure, and commitment to the defence domain.

NGO Volunteering: Supported 50+ economically disadvantaged families at a Vadodara humanitarian NGO (2024) through data entry, supply coordination, and patient intake record management.

School Leadership: School Captain & House Captain — Jawahar Navodaya Vidyalaya | **3rd Place**, Regional Science Exhibition